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From Complex Numbers to Octonions: The Structure and Evolution of Real Division Algebras

Henrique Ferreira Lourenço[a] **and Bruno Leonardo Macedo Ferreira**[a,b]✉

[a] *Federal University of Technology-Paraná-Brazil.*
[b] *University of São Paulo, Brazil.*

✉ Corresponding author: brunoferreira@utfpr.edu.br
E-mail: henriquelourenco@alunos.utfpr.edu.br

Abstract. This paper presents a historical and structural overview of the successive extensions of the real numbers leading to the octonions. We begin with the emergence of complex numbers as algebraic tools for solving polynomial equations and discuss their geometric and physical significance. We then explain the impossibility of constructing a three-dimensional division algebra extending the real numbers, which naturally motivates Hamilton's discovery of the quaternions, the first noncommutative real division algebra. Finally, we introduce the octonions as the last real normed division algebra, emphasizing their nonassociative yet alternative structure, their representation via the Fano plane, and their role within the Cayley–Dickson construction.

Keywords: Octonions; Quaternions; Complex Numbers; Real Normed Division Algebras; Cayley-Dickson Construction; Fano Plane; Exceptional Lie Groups.
AMS Subject Classification: 01A55; 15A66; 17A35; 17D05.

1 From Algebra to the Invention of Imaginary Numbers

Mathematics arose as an instrument to quantify the world, measure land, predict celestial movements, and regulate commerce. However, a problem once deemed impossible led to the separation of algebra from geometry and the acceptance of new numbers, called *imaginary*. Nearly 400 years later, these numbers would occupy a central position in fundamental theories of physics.

In 1494, Luca Pacioli, Leonardo da Vinci's teacher, published the *Summa de Arithmetica*, a work that compiled the mathematical knowledge of Renaissance Italy. Among its topics was the cubic equation, now written as:

$$ax^3 + bx^2 + cx + d = 0.$$

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Several civilizations like the Babylonian, Greek, Chinese, Indian, Egyptian, and Persian, had attempted for millennia to find a general solution to the cubic equation, without success. Pacioli concluded that such a solution was impossible.

The statement was surprising, since removing the cubic term reduces the problem to a quadratic equation, whose solution had been known since antiquity. The general solution of a quadratic equation is given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

In earlier times, the solution of such equations was achieved through geometric arguments. For example, consider the equation:

$$x^2 + 26x - 27 = 0.$$

The term x^2 was interpreted as the area of a square with side x, and the term $26x$ as the area of a rectangle with sides 26 and x. The sum of these areas equaled 27. To solve it, one used the method of completing the square:

$$x^2 + 26x + 13^2 = 27 + 13^2,$$

$$(x + 13)^2 = 196,$$

$$\sqrt{196} = 14 \quad \Rightarrow \quad x = 1 \text{ or } x = -27.$$

At that time, negative solutions were disregarded, as they lacked geometric interpretation in terms of lengths, areas, or volumes.

This limitation led to the formulation of different types of equations, always with positive coefficients. The same approach was applied to cubics. In the 11th century, Omar Khayyam classified 19 distinct cubic equations, maintaining all coefficients positive. Some approximate solutions were found through the intersection of curves such as parabolas, hyperbolas, and circles, but a general solution remained elusive.

In the early 16th century, Scipione del Ferro discovered a method to solve depressed cubics (without the quadratic term). His knowledge was passed to his student Antonio Fior, who in 1535 challenged Niccolò Tartaglia. The latter succeeded in solving all 30 proposed equations, generalizing the method of completing the square to three dimensions.

For instance, for the equation:

$$x^3 + 9x - 26 = 0,$$

Tartaglia showed that:

$$x = 2$$

is a solution.

The mathematical duel. This historical episode illustrates the culture of *mathematical duels* common at the time: del Ferro had kept his method secret, revealing it only to Fior on his deathbed. Tartaglia, in turn, condensed his strategy into an algorithm that avoided reconstructing the geometry for each problem.

Later, Gerolamo Cardano noticed that any general cubic could be reduced to a depressed cubic by the substitution:

$$x \mapsto x - \frac{b}{3a}.$$

This transformation eliminated the quadratic term. However, in some cases, solutions involved square roots of negative numbers, such as:

$$\sqrt{-1}.$$

The *casus irreducibilis*. When applied mechanically, the formula could yield expressions containing $\sqrt{-1}$ even when obvious real roots existed — as in $x^3 = 15x+4$, whose solution $x = 4$ is immediate. This paradox — the *casus irreducibilis* — exposed the tension between geometric proof and symbolic algebra: while geometry rejected "negative areas," algebra required radicals of negative numbers as intermediate steps.

Only in the 16th century did a hydraulic engineer discover how to manipulate these negative radicals so that they canceled out. This was Rafael Bombelli, born in Bologna, Italy, in 1526. Bombelli established eight rules for operating with this new quantity, now known as the imaginary unit i. Thus were consolidated the complex numbers:

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

From symbolism to complex analysis. In the following centuries, François Viète introduced symbolic algebraic notation, René Descartes popularized the term "imaginary numbers," and Leonhard Euler represented them by the letter i. Euler also explored the exponential function in depth, extending it to the complex domain and revealing the fundamental analytical relationship:

$$e^{ix} = \cos(x) + i\sin(x).$$

Among its particular cases, the most famous is $x = \pi$, yielding Euler's *identity*, which connects the most fundamental constants of mathematics:

$$e^{i\pi} + 1 = 0.$$

This identity elegantly expresses the connection between exponential and trigonometric functions. The complex exponential function describes circular motion in the complex plane, where the real and imaginary parts form continuous patterns equivalent to sine and cosine waves — the same patterns underlying physical phenomena such as sound, light, and signal processing in technology.

Euler also contributed to geometry. He demonstrated that for any polyhedron — a three-dimensional figure with flat faces, straight edges, and vertices — the following relation holds:

$$V - E + F = 2,$$

where V is the number of vertices, E the number of edges, and F the number of faces.

In the 20th century, complex numbers became central to quantum physics. In 1925, Erwin Schrödinger formulated his fundamental equation for the wave function Ψ:

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi,$$

where $\hat{H}$ represents the Hamiltonian operator.

The imaginary factor i plays a crucial role, linking the equation to the wave nature of particles. Euler's identity makes explicit the relationship between exponentials and trigonometric functions:

$$e^{ix} = \cos(x) + i\sin(x).$$

Thus, a wave function can be expressed as:

$$\Psi(x,t) = e^{i(kx-\omega t)}.$$

This formulation ensures that derivatives with respect to x or t are proportional to the function itself, providing consistency and simplicity in describing quantum waves.

Geometric interpretation of i. In the complex plane, multiplication by i corresponds to a rotation of 90°. This structural feature explains why solutions of the form $e^{i(kx-\omega t)}$ naturally arise in linear wave equations. As Freeman Dyson remarked, by introducing $\sqrt{-1}$ into Schrödinger's equation, "suddenly everything made sense": the wave description of the atom — the foundation of modern chemistry — became possible.

Thus, it is clear that imaginary numbers, initially introduced as an auxiliary tool for solving cubic equations, became indispensable for understanding the reality of our world.

2 The Impossibility of a Three-Dimensional Complex Number

When studying complex numbers, it is common to ask the following question:

> "Is it possible to define a three-dimensional numerical system that constitutes a field, with a well-defined multiplication?"

The answer is negative, and its justification can be reached in a simple way.

2.1 The case of complex numbers

Complex numbers have the form $a + bi$, with $a, b \in \mathbb{R}$, and satisfy $i^2 = -1$. Thus, the set $\mathbb{C}$ can be understood as a two-dimensional extension of $\mathbb{R}$ that indeed forms a field.

2.2 Attempt to construct a three-dimensional system

Suppose one wishes to define a field in $\mathbb{R}^3$. Let us represent a generic element as

$$(\alpha, \beta, \gamma) = \alpha + \beta i + \gamma j,$$

where $1 = (1,0,0)$, $i = (0,1,0)$, and $j = (0,0,1)$.

Assuming that there exists a well-defined multiplication that makes this set a field, the product ij must be expressible as a linear combination:

$$ij = \alpha_0 + \beta_0 i + \gamma_0 j, \quad \alpha_0, \beta_0, \gamma_0 \in \mathbb{R}.$$

Multiplying on the left by i, we obtain:

$$-j = i^2 j = i(ij).$$

Substituting the previous expression for ij, we get:

$$-j = i(\alpha_0 + \beta_0 i + \gamma_0 j) = \alpha_0 i - \beta_0 + \gamma_0 ij.$$

Applying $ij = \alpha_0 + \beta_0 i + \gamma_0 j$ again, we find

$$(\alpha_0\gamma_0 - \beta_0) + (\alpha_0 + \beta_0\gamma_0)i + (\gamma_0^2 + 1)j = 0.$$

Since 1, i, and j are linearly independent, this equality is possible only if

$$\gamma_0^2 + 1 = 0,$$

which has no solution in $\mathbb{R}$. Therefore, a contradiction arises.

The contradiction follows solely from the properties of associativity and distributivity. Even relaxing these conditions leads to new issues: the division operation may fail to exist.

For instance, assuming again $ij = \alpha_0 + \beta_0 i + \gamma_0 j$, we have

$$(i - \gamma_0)\Big[(\alpha_0\gamma_0 - \beta_0) + (\beta_0\gamma_0 + \alpha_0)i + (\gamma_0^2 + 1)j\Big] = 0,$$

with both factors nonzero, implying the existence of zero divisors and, therefore, preventing the structure from being a field.

Analyzing this simple demonstration, it is easy to see the reason for the impossibility of a three-dimensional complex number. After the complex numbers ($\mathbb{C}$, dimension 2), the next consistent system is that of the quaternions ($\mathbb{H}$, dimension 4), which, although not forming a field, constitute a noncommutative division algebra.

3 Quaternions: the First Noncommutative Algebra

3.1 Historical Origin and Geometric Motivation

The history of quaternions began with William Rowan Hamilton's search for a generalization of complex numbers capable of describing rotations in three-dimensional space. Since the 17th century, imaginary numbers had been accepted as extensions of the reals, and their geometric representation by Argand and Gauss clearly showed the relationship between multiplication and rotation in the plane. Hamilton, however, sought a similar system that could represent rotations in three-dimensional space — something impossible for complex numbers, since they operate only in two dimensions, as previously shown.

For more than a decade, Hamilton attempted to extend the idea of ordered pairs (a, b), used for complex numbers, to triples (a, b, c) that could operate consistently.

However, no rule of multiplication preserved the desired property that the modulus of the product equals the product of the moduli. On October 16, 1843, while walking with his wife along Dublin's Royal Canal, Hamilton had the decisive insight: for the algebra to work, it was necessary to introduce three mutually perpendicular imaginary units, i, j, and k, obeying the relations

$$i^2 = j^2 = k^2 = ijk = -1. \tag{1}$$

Excited, he carved this formula into the stone of Broom Bridge (now marked by a commemorative plaque). Thus was born the first noncommutative algebra in history: the quaternions.

3.2 Definition and Fundamental Algebraic Properties

The set of quaternions, denoted by $\mathbb{H}$, is defined as the four-dimensional real vector space

$$\mathbb{H} = \mathrm{span}_{\mathbb{R}}\{1, i, j, k\},$$

endowed with the fundamental relations

$$i^2 = j^2 = k^2 = ijk = -1, \qquad ij = k = -ji, \tag{2}$$

and their cyclic permutations.

Every element $q \in \mathbb{H}$ can be uniquely written as

$$q = a + bi + cj + dk, \quad a, b, c, d \in \mathbb{R}. \tag{3}$$

The product of two quaternions $p = a_1 + b_1 i + c_1 j + d_1 k$ and $q = a_2 + b_2 i + c_2 j + d_2 k$ is given by

$$\begin{aligned} pq = {} & (a_1a_2 - b_1b_2 - c_1c_2 - d_1d_2) \\ & + (a_1b_2 + a_2b_1 + c_1d_2 - c_2d_1)i \\ & + (a_1c_2 - b_1d_2 + c_1a_2 + d_1b_2)j \\ & + (a_1d_2 + b_1c_2 - c_1b_2 + d_1a_2)k. \end{aligned} \tag{4}$$

The *conjugate* of q is defined as

$$\bar{q} = a - bi - cj - dk, \tag{5}$$

and its *norm* as

$$|q|^2 = q\bar{q} = a^2 + b^2 + c^2 + d^2. \tag{6}$$

From this definition, it follows immediately that $\mathbb{H}$ is an associative division algebra over $\mathbb{R}$, since every nonzero element has an inverse:

$$q^{-1} = \frac{\bar{q}}{|q|^2}. \tag{7}$$

These properties show that $\mathbb{H}$ preserves the multiplicative norm $|pq| = |p||q|$, making it a *composition algebra*. On the other hand, since $ij \neq ji$, it is not commutative — an unusual and revolutionary fact in the history of algebra.

3.3 Matrix Model and Relation to Complex Numbers

Just as complex numbers can be represented by real 2×2 matrices, quaternions admit a complex matrix representation. Let $z, w \in \mathbb{C}$, and define the set

$$\mathbb{H} = \left\{ \begin{pmatrix} z & -w \\ \bar{w} & \bar{z} \end{pmatrix} : z, w \in \mathbb{C} \right\}.$$

This representation is an $\mathbb{R}$-algebra isomorphism, preserving addition and multiplication, and provides a practical way to perform computations in $\mathbb{H}$. The determinant of the matrix associated with q equals its norm $|q|^2$, and conjugation corresponds to the Hermitian transpose.

3.4 Geometric Interpretation and Rotations in Space

Analogous to complex numbers, a quaternion can be decomposed into a *real part* and an *imaginary part*:

$$q = a + \mathbf{v}, \qquad \mathbf{v} = bi + cj + dk. \tag{8}$$

The subset $\mathrm{Im}(\mathbb{H}) = \{bi + cj + dk : b, c, d \in \mathbb{R}\}$ is called the *imaginary space* of quaternions and is isomorphic to $\mathbb{R}^3$.

For a vector $\mathbf{v} \in \mathbb{R}^3$, identified with a purely imaginary quaternion, and a unit quaternion

$$q = \cos\left(\frac{\theta}{2}\right) + \mathbf{n}\sin\left(\frac{\theta}{2}\right),$$

where $\mathbf{n}$ is a unit vector, the transformation

$$\mathbf{v}' = q\,\mathbf{v}\,\bar{q} \tag{9}$$

represents a rotation of $\mathbf{v}$ around the axis $\mathbf{n}$ by an angle θ.

This property makes quaternions extremely useful tools for describing rotations in three dimensions, free from the ambiguities of Euler coordinates. In computer graphics and robotics, this formulation is widely used to represent orientations and the motion of rigid bodies.

3.5 Dot Product, Cross Product, and Lie Structure

The multiplication of two purely imaginary quaternions naturally combines the usual dot and cross products. If $\mathbf{u}, \mathbf{v} \in \mathrm{Im}(\mathbb{H})$, then:

$$\mathbf{uv} = -\langle \mathbf{u}, \mathbf{v} \rangle + \mathbf{u} \times \mathbf{v}, \tag{10}$$

where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product and $\times$ the cross product in $\mathbb{R}^3$. The anticommutativity $\mathbf{uv} = -\mathbf{vu}$ implies that $(\mathrm{Im}(\mathbb{H}), \times)$ is a Lie algebra, satisfying the Jacobi identity:

$$\mathbf{u} \times (\mathbf{v} \times \mathbf{w}) + \mathbf{v} \times (\mathbf{w} \times \mathbf{u}) + \mathbf{w} \times (\mathbf{u} \times \mathbf{v}) = 0.$$

3.6 Legacy and Influence of Quaternions

Hamilton saw in his creation a new language for geometry and physics, predicting that it would be widely used to describe natural phenomena. Decades later, his predictions were partially confirmed: James Clerk Maxwell incorporated quaternionic ideas into the formalism of electromagnetism, and Oliver Heaviside and J. Willard Gibbs derived from these structures the modern vector calculus, with the operations of divergence and curl.

Even today, quaternions are essential in many areas, from quantum mechanics to 3D animation. Thus, it is undeniable that quaternions represent a milestone in the transition between algebra and geometry.

4 The Octonions: Structure, History, and Fundamental Properties

4.1 Historical Introduction

The search for numerical systems that generalized the complex numbers profoundly influenced the development of algebra in the nineteenth century. After the creation of the quaternions by William Rowan Hamilton in 1843, a new phase of mathematics began—one focused on expanding numerical structures and understanding the geometric interpretations associated with each extension. In this context, the octonions emerged: an eight-dimensional algebra that preserves the division property but definitively breaks associativity.

John Thomas Graves, a friend and collaborator of Hamilton, was the first to describe such a structure, which he called *octaves*. His discovery occurred only a few months after the emergence of the quaternions, in December 1843, when Graves realized that it was possible to create an even broader algebra obtained by doubling the quaternions. In his correspondence with Hamilton, Graves demonstrated that this new construction possessed a multiplicative norm, granting it the property of being a division algebra. However, his work was not immediately published. Only in 1845 did Arthur Cayley present a paper containing equivalent results, which led the octonions to become known as the *Cayley numbers*.

For decades, the octonions remained largely ignored by the mathematical community. The absence of immediate geometric applications and the fact that they were non-associative led many to view them as an algebraic curiosity. It was only in the early twentieth century, with Élie Cartan's studies on exceptional groups and the phenomenon of *triality*, that interest in the octonions reemerged. Cartan showed that the octonionic algebra is intrinsically connected to the symmetry between vectors and spinors in eight-dimensional Euclidean spaces, establishing a bridge between algebra and differential geometry. Later, the 1934 research of Jordan, von Neumann, and Wigner revealed that the octonions also play an important role in the formalism of Jordan algebras, which arose in the context of quantum mechanics.

Throughout the twentieth century, new connections were identified: the octonions are related to the construction of exceptional algebras such as $J_3(\mathbb{O})$, to the structure of the Lie group G_2, and to unifying theories of modern physics, including versions of string theory. These developments consolidated the status of the octonions as the last and most complex of the real normed division algebras.

4.2 Construction and Algebraic Notation

The octonions, denoted by $\mathbb{O}$, form a real eight-dimensional algebra. Their basis is traditionally written as

$$\{1, e_1, e_2, e_3, e_4, e_5, e_6, e_7\},$$

where 1 is the multiplicative identity and each e_i is an imaginary unit satisfying $e_i^2 = -1$.

The product between these units is neither commutative nor associative, but it follows an oriented multiplication pattern described by the *Fano Plane*, schematically represented in Figure 1. This diagram consists of seven points and seven lines (with the circle also counted as a line connecting three points), where each line contains exactly three imaginary units, and each triple (e_i, e_j, e_k) follows a cyclic relation:

$$e_i e_j = e_k, \quad e_j e_i = -e_k.$$

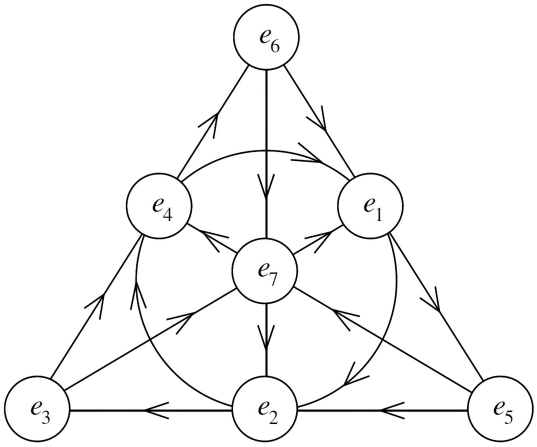


Fig. 1. Fano Plane: mnemonic representation of the multiplication among the imaginary units of the octonions.

This organization allows one to reconstruct the entire multiplication table of $\mathbb{O}$ without memorizing all combinations. Despite their complexity, the octonions maintain properties that make them remarkably structured: they are an *alternative algebra*, meaning that the associator $(x, y, z) = (xy)z - x(yz)$ vanishes whenever two of the elements coincide, that is, $(x, x, y) = (y, x, x) = 0$.

Parts of the exposition and background material concerning alternative algebras in this article follow the treatment developed in Ferreira [9]. Readers interested in a more comprehensive introduction to alternative algebras, including further examples, identities, and structural results, are referred to that book for additional details.

4.3 Fundamental Operations

Let $x, y \in \mathbb{O}$, with

$$x = x_0 + \sum_{i=1}^{7} x_i e_i, \quad y = y_0 + \sum_{i=1}^{7} y_i e_i, \quad x_i, y_i \in \mathbb{R}.$$

The *conjugate* of x is defined as

$$\bar{x} = x_0 - \sum_{i=1}^{7} x_i e_i,$$

and the associated *norm* as

$$\|x\| = \sqrt{x\bar{x}} = \sqrt{x_0^2 + x_1^2 + \cdots + x_7^2}.$$

Multiplication preserves this norm, so that

$$\|xy\| = \|x\| \, \|y\|,$$

a property that ensures $\mathbb{O}$ is a division algebra: every nonzero element has an inverse given by

$$x^{-1} = \frac{\bar{x}}{\|x\|^2}.$$

However, unlike the quaternions, one cannot freely reorder the factors in products of three or more terms, since associativity generally fails.

4.4 The Octonions and the Cayley–Dickson Process

From a constructive point of view, the octonions can be obtained from the quaternions through the Cayley–Dickson process. The central idea is to double a known algebra, creating pairs (a, b) with a product defined by

$$(a, b)(c, d) = (ac - d\bar{b},\ \bar{a}d + cb),$$

where the conjugation is inherited from the previous algebra. Applying this rule successively, we obtain the sequence

$$\mathbb{R} \longrightarrow \mathbb{C} \longrightarrow \mathbb{H} \longrightarrow \mathbb{O}.$$

Each step doubles the dimension, and beyond the octonions the structure loses both alternativity and division, giving rise to the so-called sedenions and higher stages of the construction. The octonionic algebra is therefore the last level of the process in which a multiplicative norm and well-defined inverses still exist.

4.5 Algebraic and Geometric Properties

Although non-associative, the octonions preserve a weaker form of algebraic consistency called *alternativity*. This means that the associator $(x, y, z) = (xy)z - x(yz)$ vanishes whenever two of the elements coincide, i.e.,

$$(x, x, y) = (y, x, x) = 0 \quad \text{for all } x, y \in \mathbb{O}.$$

From alternativeity follow the Moufang identities, such as

$$(xy)(zx) = x(yz)x,$$

which ensure a certain internal coherence of the product. These identities are fundamental in the study of *Moufang geometries* and the so-called *simple alternative algebras*.

The automorphism group of the octonion algebra is the exceptional Lie group G_2, which preserves the multiplicative structure of $\mathbb{O}$. This group is associated with remarkable geometric properties, such as the parallelizability of the seven-dimensional sphere and the existence of invariant differential forms, which play a central role in G_2-holonomy geometry and its applications in theoretical physics.

Historical and Cultural Notes

The suffix *–nion*, introduced by Hamilton in the term *quaternion* (from the Latin *quaterni*, "in groups of four"), was retained in subsequent extensions of the Cayley–Dickson process. Thus, terms such as *octonion* and *sedenion* consolidated a phonetic tradition that associates the number of dimensions with the corresponding Greek or Latin numerical root, followed by the ending *–nion*. This convention eventually came to characterize the entire family of hypercomplex algebras obtained by successive dimension doubling.

However, beyond 16 dimensions, the traditional Latin-based nomenclature—such as *trigintaduonion* (32D), *sexagintaquattuornion* (64D), *centumduodetrigintanion* (128D), and *ducentiquinquagintasexion* (256D)—became excessively long and impractical. With the advancement of computational studies and high-dimensional visualizations, shorter and more symbolic alternative denominations began to emerge.

Robert P. C. de Marrais and Tony Smith proposed an alternative nomenclature, later adopted by Michael Carter in his visualizations at the *Maple Application Center*. This new sequence was conceived both for reasons of phonetic simplicity and symbolic associations with cultural and mathematical systems.
The first nine hypercomplex algebras arising from the Cayley–Dickson process can be listed as follows:

$\mathbb{R}$ (1D), $\mathbb{C}$ (2D), $\mathbb{H}$ (4D), $\mathbb{O}$ (8D), $\mathbb{S}$ (16D), $\mathbb{P}$ (32D), $\mathbb{X}$ (64D), $\mathbb{U}$ (128D), $\mathbb{V}$ (256D).

or, respectively, named: reals, complexes, quaternions, octonions, sedenions, pathions, chingons, routons, and voudons.

The adoption of short names for higher-dimensional algebras was not limited to a matter of phonetic convenience. Beyond linguistic economy, authors such as Robert P. C. de Marrais and Michael Carter sought to assign cultural and structural meanings to the higher stages of the Cayley–Dickson process. These proposals connect the dimensions 32, 64, 128, and 256 to traditional systems of knowledge organization and historical references, thereby endowing hyperdimensions with a conceptual character that transcends mere binary counting. Such associations also function as metaphors for the very mechanism of recursive doubling that characterizes the Cayley–Dickson process, allowing each dimensional leap to be visualized as a combinatorial reorganization analogous to those found in multicultural symbolic systems.



Fig. 2. Sedenion (16D) — composition of contravariant figures.

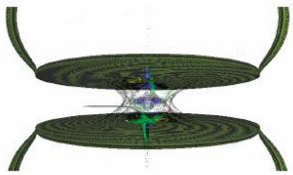


Fig. 3. Pathion (32D) — contravariant and covariant structure.



Fig. 4. Chingon (64D) — contravariant nybbles.

Fig. 5. Visual evolution of the Cayley–Dickson hypercomplex algebras, according to Michael Carter (2011). The figures illustrate the increase in complexity and symmetry along the doubling process, from the *Sedenions* (16D) to the *Pathions* (32D) and *Chingons* (64D). Source: adapted from Carter, M. *Visualization of the Cayley–Dickson Hypercomplex Numbers Up to the Chingons (64D). MaplePrimes*, 2011.

The *Pathions* (32 dimensions) derive their name from the "32 Paths of Wisdom" of the Jewish Kabbalistic tradition. These paths express the connections between the ten spheres of the Tree of Life (*Sefirot*) and the 22 letters of the Hebrew alphabet, producing a structure of $10 + 22 = 32$ fundamental elements. In Kabbalah, these 32 paths represent internal relationships among components that combine to form a symbolic totality. This logic closely parallels the dimensional expansion observed in the pathions: at each stage of construction, new interactions arise between pairs of components, in a manner analogous to the way Kabbalah organizes its paths as a network of interdependent connections. Thus, the analogy transcends a purely numerical reference and suggests a structural affinity with the multiplicity of internal relations present in the 32-dimensional algebra.

The *Chingons* (64 dimensions) are inspired by the 64 hexagrams of the *I Ching*, the classical Chinese philosophical text. Each hexagram consists of six binary lines (yin/yang), forming 2^6 possible combinations—exactly 64 distinct states. The association with the Cayley–Dickson process is particularly deep, as both are grounded in successive binary constructions. Each hexagram may be interpreted as a six-bit word composed of complementary opposites; analogously, the transition to 64 dimensions in the hypercomplex construction results from the systematic doubling of coordinates and the interaction between pairs of components. The combinatorial structure of the *I Ching*, organized as a catalog of possible states and transformations, echoes the internal geometry of the chingons, where successive binary combinations generate a vast, yet systematically structured space.

The *Routons* (128 dimensions) refer to Route 128, the highway encircling Boston that became, between the 1950s and 1980s, one of the first modern technological hubs. Although seemingly less symbolic, this association carries a significant structural metaphor: the region represented a landmark of exponential technological growth, intensive connectivity, and the development of complex computational systems—features directly related to the handling and visualization of hypercomplex

structures of high dimensionality. Moreover, the notion of "routing" embedded in the name evokes the internal organization of the routons, whose multiplication involves distributed information flows governed by highly structured patterns, comparable to advanced computational processes that emerged in this historical context.

The *Voudons* (256 dimensions), in turn, reference to the African divinatory system Ifá, which recognizes 256 *odù*. Each *odù* corresponds to a combination produced by four binary pairs (eight bits), yielding a total of $2^8 = 256$ possible archetypes. Unlike the other symbolic systems mentioned, Ifá is explicitly organized as a complete and exhaustive binary system, forming a closed combinatorial cycle. This feature aligns closely with Carter's observation that hypercomplex structures beyond 256 dimensions tend to exhibit repetitive or saturated geometric patterns in computational visualizations. Thus, dimension 256 simultaneously represents a combinatorial closure within the Ifá system and an empirical saturation point in the visualizations of the voudons, reinforcing the structural coherence of this choice.

According to Carter, although algebraic doubling may continue *ad infinitum*, from the standpoint of computational visualization, one observes a repetition of patterns beyond approximately 1024 dimensions, at which point visually distinct geometric properties cease to emerge. Furthermore, implementations in *Maple* are typically limited to 256 dimensions due to the prohibitive computational cost of operating with matrices and tensors of such magnitude. Although these denominations are not part of the classical mathematical nomenclature, they have gained traction in niches of computational visualization, experimental algebraic theory, and informal studies in mathematical physics, acting as mnemonic labels that combine historical tradition, binary combinatorics, and algebraic abstraction.

References

[1] J. C. Baez, The octonions, *Bull. Amer. Math. Soc.*, **39** (2002), 145–205.

[2] M. T. Carter, The Cayley–Dickson algebra from 4D to 256D, *Maple Application Center*, 2010.

[3] M. T. Carter, Visualization of the Cayley–Dickson hypercomplex numbers up to the chingons (64D), *MaplePrimes*, 2011.

[4] A. Elduque, *Flexible Composition and Okubo Algebras* (expository slides/notes).

[5] A. Hurwitz, Über die Composition der quadratischen Formen, *Nachr. Ges. Wiss. Göttingen*, (1898), 309–316.

[6] L. A. Peresi, *Álgebras Não-Associativas* (course notes), IME–USP.

[7] R. D. Schafer, *An Introduction to Nonassociative Algebras*, Academic Press, 1966.

[8] M. L. Tomber, A short history of nonassociative algebras, *Hadronic Journal*, **2** (1979), 1252–1387.

[9] B. L. M. Ferreira, *Uma Breve Introdução às Álgebras Alternativas*, Bukhtishu Press, Sofia, 2026. doi: 10.64721/9786199398623.